Higher National Diploma in Information Technology
Second Year First Semester Examination – 2022
HNDIT 3032 – Data Structures and Algorithms

Instructions:
Answer any four (04) questions
Every question carry 25 marks

No. of questions: 05
No. of pages : 03
Time : Two hours

Question 01 **(Total Marks 25)**

i) What is the Data Structure? (03 Marks)

ii) What is ADT? Give two examples of ADT (2 + 2 = 04 Marks)

iii) Write a Java method to check the User input string is a **palindrome or Not using a loop.** (Hint: **palindrome word is** civic) (07 Marks)

iv.) Write a Java method to **find Maximum Number** in 10 **unsigned integers** an Array (07 Marks)

v.) Find the **Big O** of the given equation $F(n)=n^3+4n^2+5$ (04 Marks)

Question 02 **(Total Marks 25)**

i.) Consider following Java code and Write a Java method to add two metrics **(metricsA , metricsB)** and display it as a Metrics .

```
int metricsA[][]={{10,12,13},{11,15,16},{20,21,23}};
int metricsB[][]= {{14,9,8},{1,2,3},{60,71,73}};
```

(08 Marks)

ii.) Compare and Contrast LINKED List Vs an Array (04 Marks)

iii.) Write Four Operation in LINKED LIST (04 Marks)

iv.) Write a Java code to define a Linked List data structure using Java Classes (05 Marks)

v.) Write a pseudo code appending Node in LINKED LIST (04 Marks)

Question 03 **(Total Marks 25)**

i.) Consider following Java coding in Queue implementation Java Program.

```
public class ArrayQuequ {
        public int size;
        int front,rear;
        public int que[]=new int[size];
        public ArrayQuequ(int size, int[] arr) {
                this.size = size;
                this.que = arr;
                front=-1;
                rear=-1;}
        public int getSize() {
                return size;}
        public int[] getQue() {
                return que;}
```

a.) Write a reason why take two variables **rear** and **front** implementing Queue data structure. (04 Marks)

b.) Write Java Method to represent enqueue operation in a queue (04 Marks)

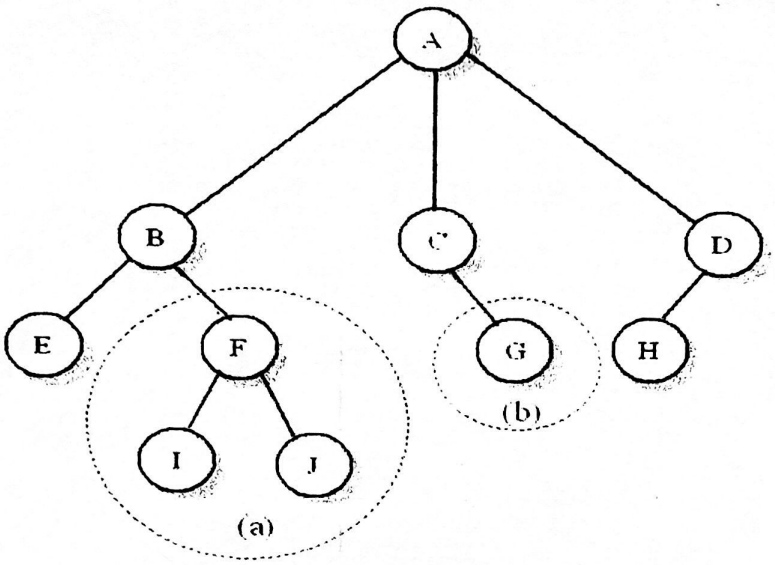
c.) Briefly explain why circular queue representation is introduced. (05 Marks)

ii) What is STACK? Discuss the LIFO behavior of a STACK. (03 Marks)

iii.) Graphically illustrate Following operation in STACK implementation, its **size is 4**

a.)Empty STACK
b.)x=isEmpty();
c.)push(10);
d.)Push(12);
e.)push(15);
f)y=pop();
g.)push(90);
h.)push(100);
j.)z=isFull(); (09 Marks)

Question 04 **(Total Marks 25)**



i.) Consider the above diagram

a.) Define the parent node. Give an example. (02 Marks)
b.) What is **Height** of a Tree (02 Marks)
c.) Define the leaf nodes. Give two examples. (03 Marks)
d.) Write a Path From **root to J** (02 Marks)

ii.) Define the Binary Search Tree (BST) (04 Marks)

iii.)Insert given Number Set in to **BST** (23,33,45,30,10,55,15,8,42,17) (05 Marks)

iv.) What is **Selection short** (02 Marks)

v.) Mentions the steps to Sort the given numbers 6, 7,72, 4, 32, 65, 9, 56 using **Bubble sort** (05 Marks)

Question 05 **(Total Marks 25)**

i.) What is search algorithm? Briefly explain the two types of searching Algorithms (2+2*2 = 06 Marks)

ii.) Write a Java code to Sequential/Linear Search (05 Marks)

iii.) Write a Java program to search 99 in a following Array using Sequential/Linear Search
int num[] = {4, 65, 2, -31, 0, 99, 2, 83, 782, 1}; (05 Marks)

iv.) Discuss Efficiency of Sequential/Linear Search and Binary search algorithm (04 Marks)

v.) **"Selection sort algorithm** is better than **bubble sort algorithm"**. Do you agree? Justify your answer (05 Marks)